

St. Thomas' College of Engineering and Technology
Final Year Eighth Semester Synopsis

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Matrimonial Matchmaking: A Machine Learning Approach

Prepared by

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Project Synopsis

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Declaration

We are submitting the synopsis in seventh semester on “Matrimonial Matchmaking: A Machine Learning Approach” as a part of our final year project under the guidance of Dr. Arindam Chakravorty.

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Vision & Mission of the Institute

Vision: To evolve as an industry oriented, research-based Institution with ethical values for creative solutions in various engineering domains, with an ultimate objective of meeting technological challenges faced by the Nation and the Society.

Mission:

- To enhance the quality of engineering education and delivery through accessible, comprehensive and research-oriented teaching-learning-assessment processes in the state-of-art environment.
- To create opportunities for students and faculty members to acquire professional knowledge with multidisciplinary approach and develop managerial, entrepreneurial and social attitudes with highly ethical and moral values.
- To satisfy the ever-changing needs of the nation with respect to evolution and absorption of sustainable and environment friendly technologies for effective creation of knowledge-based society in the global era.

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Vision, Mission, PEO & PSO of the Department

Vision: To promote the advancement of learning in Information Technology through research-oriented dissemination of knowledge with ethical values which will lead to innovative applications of information in industry and society.

Mission:

- To incubate students grow into industry ready professionals, proficient research scholars and enterprising entrepreneurs.
- To create a learner- centric and sustainable environment that motivates the students in adopting emerging technologies of the rapidly changing information society.
- To foster innovative responsiveness to social, environmental, and technological challenges among faculty and students.

PEO: Graduates of Information Technology Program shall

PEO1: Excel in professional career, higher education, and research with a strong foundation in core and allied areas of Information Technology.

PEO2: Demonstrate professionalism, entrepreneurship, ethical behaviour, effective communication, and collaborative teamwork to adapt to emerging trends through lifelong learning.

PEO3: Exhibit the skills and knowledge required to design, develop, and implement sustainable and ethically sound IT solutions by adopting multidisciplinary approach to solve real-life problems.

PSO: On completion of the program the graduates will be able to

PSO1 (Professional Skill): Apply their knowledge in the field of information technology and contribute significantly to the corporate world by way of providing appropriate solutions to engineering problems and establish their skills in high performance computing, software engineering, programming and thrust areas like security and machine intelligence.

PSO2 (Competency): Demonstrate their proficiency in analytical and critical thinking, methodologies of practical design, data analysis and interpretation through their technical expertise which will help them to excel in higher studies within the country and abroad.

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Project Mapping with PO & PSO

Project Mapping with Program Outcomes

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
3	3	3	3	3	2	2	2	3	2	3

Justification:

PO1 (Level 3): Uses math, statistics, and ML concepts to build the matchmaking system.
PO2 (Level 3): Analyzes complex matching and compatibility problems using research methods.
PO3 (Level 3): Designs a complete solution including rule-based and ML modules.
PO4 (Level 3): Conducts experiments, model testing, and data evaluation for valid results.
PO5 (Level 3): Applies modern tools like Python, ML libraries, and NLP techniques.
PO6 (Level 2): Considers cultural, social, and ethical factors in matchmaking design.
PO7 (Level 2): Ensures ethical handling of personal and sensitive user data.
PO8 (Level 2): Involves teamwork in planning, development, and execution.
PO9 (Level 3): Requires effective documentation, reporting, and presentation.
PO10 (Level 2): Uses basic project management through WBS, timelines, and task allocation.
PO11 (Level 3): Involves learning emerging ML and NLP technologies throughout the project.

Project Mapping with Program Specific Outcomes

PSO1	PSO2
3	2

Justification:

The project achieves PSO1 by applying IT and software engineering skills to develop a complete matchmaking system using machine learning, data processing, programming, and intelligent decision-making techniques. It demonstrates practical use of high-performance computing tools and machine intelligence concepts to solve a real-world engineering problem.

It also fulfills PSO2 by showcasing strong analytical thinking, data interpretation, and design methodologies through tasks such as feature engineering, model training, evaluation, and system integration—skills that support advanced studies and technical growth.

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Introduction:

The rise of digital technologies has transformed how individuals search for compatible life partners, especially in cultures where partner selection depends on personal preferences, family expectations, and social compatibility. Online matrimonial platforms now offer large pools of potential matches, yet many still rely on basic demographic filters such as age, religion, caste, education, and location. These simple filters fail to capture deeper compatibility aspects like personality, lifestyle choices, and mutual expectations, resulting in irrelevant or incomplete match suggestions.

To overcome these limitations, the proposed project—"Matrimonial Matchmaking: A Machine Learning Approach"—introduces a hybrid recommendation system that combines rule-based mutual preference matching with machine learning-based compatibility prediction. By integrating both explicit user preferences and insights learned from historical successful matches, the system aims to deliver more accurate, adaptive, and user-centric partner recommendations.

CHAPTER 1

1.1 Problem Statement: To design and develop a Machine Learning based Matrimonial Matchmaker application that will provide users with compatible, reliable matches.

1.2 Problem Definition: The project aims to design a hybrid matchmaking system that ensures mutual compatibility and improves prediction accuracy. The system must:

- I. Match users only when both individuals satisfy each other's stated preferences (age, education, co lifestyle, etc.).
- II. Enhance compatibility scoring by learning from historical successful match data using supervised ML models.
- III. Process free-text expectations using NLP to extract meaningful features.
- IV. Generate a ranked list of recommendations based on a combined rule-based and ML-derived scoring mechanism.

1.3 Objective:

The primary objective is to develop an intelligent matchmaking system using mutual preference filtering and machine learning to improve accuracy, relevance, and personalization.

- I. Design a structured data model for profiles and partner preferences.
- II. Implement rule-based mutual preference filtering to remove incompatible options.
- III. Build NLP pipelines for text-based preference processing.
- IV. Train supervised machine learning models using historical match data.

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V. Integrate rule-based and ML predictions into a unified scoring and ranking system.

VI. Provide a simple user interface for profile creation and match viewing.

1.4 Tools and Platform

The proposed Matrimonial Matchmaking System is developed using modern machine learning and software engineering technologies to ensure efficient profile processing, compatibility prediction, and recommendation generation.

Programming Language (Python)

Python is selected as the primary programming language because of its extensive support for machine learning, data analysis, and web application development. It provides a rich ecosystem of libraries that simplify data preprocessing, model training, and deployment.

Data Processing Libraries

Pandas (Used for):

- Dataset loading and manipulation
- Handling missing values
- Data cleaning and transformation
- Tabular profile management

NumPy (Used for):

- Numerical computations
- Matrix operations
- Feature vector generation
- Statistical calculations

Machine Learning Libraries

Scikit-Learn (Used for):

- Data preprocessing
- Label encoding
- Feature scaling
- Model training and evaluation

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- Performance metrics computation

XGBoost (Used for):

- Compatibility prediction
- Classification of successful and unsuccessful matches
- Feature importance analysis
- High-performance predictive modeling

Natural Language Processing (NLP)

NLTK / SpaCy (Used for):

- Text preprocessing
- Tokenization
- Stop-word removal
- Lemmatization
- Extraction of meaningful features from textual partner preferences

Database Management System

MySQL (Used for):

- User profile storage
- Preference storage
- Match history management
- Recommendation record maintenance

User Interface Development

Flask (Used for):

- Backend service implementation
- API development
- Communication between frontend and ML modules

HTML, CSS, and JavaScript (Used for):

- User profile creation forms
- Preference selection interfaces

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- Match result visualization

Development Tools

Visual Studio Code (VS Code) (Used for):

- Code development
- Debugging
- Project management

Git and GitHub (Used for):

- Version control
- Collaborative development
- Source code maintenance

Platform Requirements

Hardware Requirements

- Processor: Intel Core i5 or higher
- RAM: 8 GB minimum
- Storage: 20 GB free space

Software Requirements

- Windows 10/11 or Linux
- Python 3.11+
- MySQL Server
- VS Code
- Modern Web Browser

The combination of these tools and technologies enables the development of a scalable, efficient, and intelligent matchmaking system capable of processing large numbers of user profiles while delivering personalized partner recommendations.

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1.5 BRIEF DISCUSSION ON THE PROBLEM:

Modern matrimonial matchmaking systems face challenges because most traditional platforms rely only on simple demographic attributes such as age, location, and religion. These limited criteria cannot accurately capture deeper relationship factors like compatibility in values, long-term commitment, emotional expectations, and socio-cultural alignment.

In contrast, the problem addressed in this project focuses on developing a more data-driven and holistic matchmaking model using features such as age gap, education level, income preference, cultural preference, location. Even without personal descriptions or lifestyle text data, the challenge lies in effectively analyzing these structured attributes to recommend partners who share mutual compatibility. The goal is to overcome the limitations of traditional filtering systems by using machine learning to identify deeper relationship patterns hidden within the structured dataset.

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CHAPTER 2

CONCEPTS AND PROBLEM ANALYSIS

Designing an intelligent matrimonial matchmaking system requires a comprehensive understanding of how different computational techniques interact to evaluate user compatibility. Matchmaking is an inherently complex problem because it involves structured demographic attributes, user-defined partner preferences, subjective textual descriptions, and the need for adaptive decision-making. This section presents a detailed explanation of the concepts employed in the project—ranging from rule-based matching, feature engineering, and NLP to machine learning-driven compatibility prediction and hybrid scoring methods. Each concept is analyzed in terms of how it contributes to solving key challenges in selecting accurate and meaningful partner suggestions.

Rule-Based Mutual Preference Matching (Flat Matching)

The matchmaking process begins with a rule-based mutual preference filtering system, often referred to as “flat matching.” This step is essential because it ensures that only logically compatible profiles are considered for further processing. By applying structured rules derived from user preferences, the system eliminates unsuitable profiles early, reducing computational complexity and increasing the chance of generating meaningful matches.

The system uses two groups of features derived from the user’s profile and their stated preferences:

1. **Profile Attributes:** the characteristics of an individual.
2. **Preference Attributes:** the conditions they expect their partner to satisfy.

For every potential match, the system compares these two sets of attributes from both sides (mutual checking). These features form the foundation of both the flat (rule-based) matching and the ML-based compatibility model.

Profile Attributes	Preference Attributes
I. Age: Age of the user.	Preferred Age Range. ie. Min - Max age (String, e.g., "25–30")
II. Gender: User’s gender identity	By default opposite gender.
III. Education: User’s education qualification	Preferred Education or minimum qualification
IV. Caste: User’s caste or sub-caste	Preferred caste, required caste/sub-caste for partner
V. Profession: User’s occupation/job role	Preferred profession

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VI. Religion: User's religion	Preferred religion
VII. Residence: User's residence(city/state)	Preferred partner's location (city/state/locality)

Flat Matching Process: The flat matching process is executed in several sequential steps:

I. Data Retrieval: The system begins by loading the seeker's profile, preference profile, and priority settings. It then retrieves opposite-gender candidate profiles and applies an activity filter, ensuring that only candidates active within the last 15 days are included in the matching process.

II. Mutual Minimum Preference Check: Each candidate undergoes an initial screening to verify that at least two key preference attributes are mutually satisfied in both directions. Candidates who fail to meet this minimum requirement are immediately excluded from further evaluation.

III. Feature-Based Evaluation: For all candidates who pass the preliminary check, the system evaluates compatibility across six primary features: age, religion, caste, education, profession, and residence. The number of matching features serves as the base compatibility measure for subsequent scoring.

IV. Scoring Model Selection: If the seeker has assigned priority weights to their preferences, the system uses a weighted scoring method to compute compatibility. In the absence of priority settings, a flat scoring method is applied, converting the match count into a percentage for both directions.

V. Mutual Compatibility Calculation: The final compatibility score is computed as the average of the seeker-to-candidate and candidate-to-seeker percentages. This ensures that the final match score reflects mutual suitability rather than one-sided preference satisfaction.

VI. Ranking Found Matches: All valid candidates are ranked in descending order based on their mutual match percentage. The system then updates the database to store the seeker's match list and reciprocally updates each candidate's profile to maintain a synchronized record of compatible profiles.

The rule-based flat matching process acts as an effective first-level filter that ensures only logically compatible and recently active profiles are considered for matchmaking. By applying mutual preference checks, evaluating key features, and generating transparent compatibility scores, the system eliminates irrelevant or unsuitable candidates early. This not only improves the quality and relevance of recommendations but also reduces computational load for later

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stages, creating a strong and reliable foundation for subsequent machine-learning-based compatibility analysis

Machine Learning Layer for Enhanced Match Ranking:

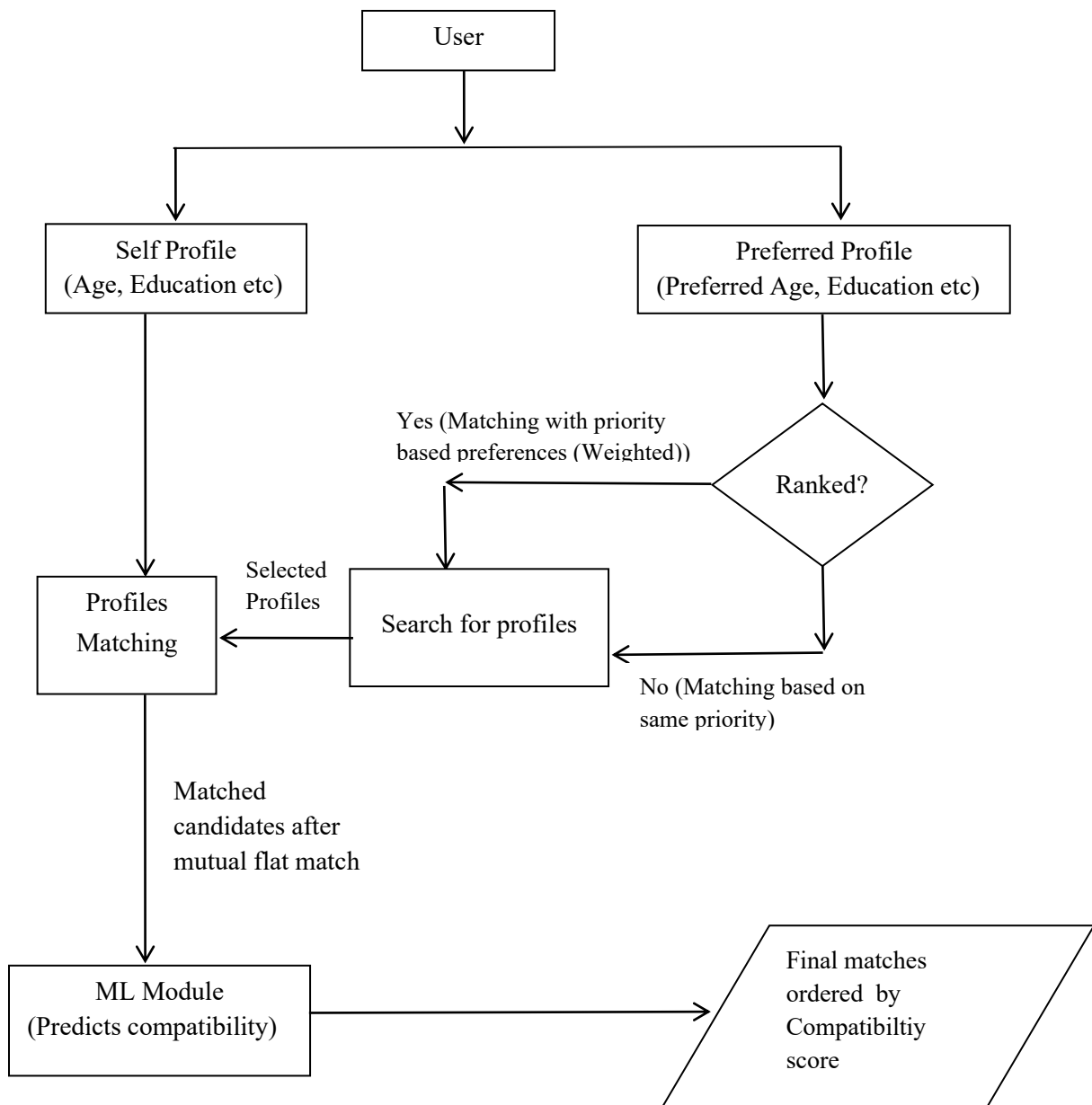
- I. **Data Preparation:** Encode the Raw Profile Data First, all the categorical, static profile information for every user (like Religion, Education_Level, and Income_Level) is converted into a numerical format. This creates a simple, structured feature vector for every profile. For a new potential match (User A and User B), you now have their clean, explicit profiles ready for the final step.
- II. **Relational Graph:** Build the Relational Graph A vast network (graph) is built from all historical users. Every user is a Node. The Edges (connections) between them are established using the Relational Features: which profiles clicked whom, who sent messages to whom, and critically, the final Marital Satisfaction/Divorce Status outcome of past paired matches. This historical behavioral data is the raw fuel for the predictive model.
- III. **Feature Generation:** The GNN Feature Generation phase takes the entire user network and runs a Graph Neural Network (GNN) to perform message passing. This is where information about who connected with whom and what the final outcome was (High_Yes or Low_No) is propagated and aggregated across the graph. This process generates the GNN Embedding: a single, dense numerical vector for every user. This vector consists of a summary of the user's hidden, behavioral preferences and their pattern of interaction with successful (or unsuccessful) matches, which is a much richer feature than their static profile alone.
- IV. **Input Creation:** Pair and Concatenate the Vectors When a potential match between User A and User B is requested, the system retrieves three key pieces of data and joins them together:
 - 1) User A's GNN Embedding (the latent behavioral score).
 - 2) User B's GNN Embedding (their latent behavioral score).
 - 3) The Combined Profile Features (the comparison of their explicit data, e.g., is their Caste_Match Same or Different).
- V. **Final Prediction:** The Classifier Calculates the Score The final Deep Learning Classifier (such as an XGBoost model) is now fed this comprehensive Input Vector. This model has been trained to map similar input patterns to the known High_Yes or Low_No outcomes.

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The classifier processes all these features—both the static profile data and the deep GNN behavioral features—and outputs the final Compatibility Score, which is a probability of achieving high marital satisfaction.

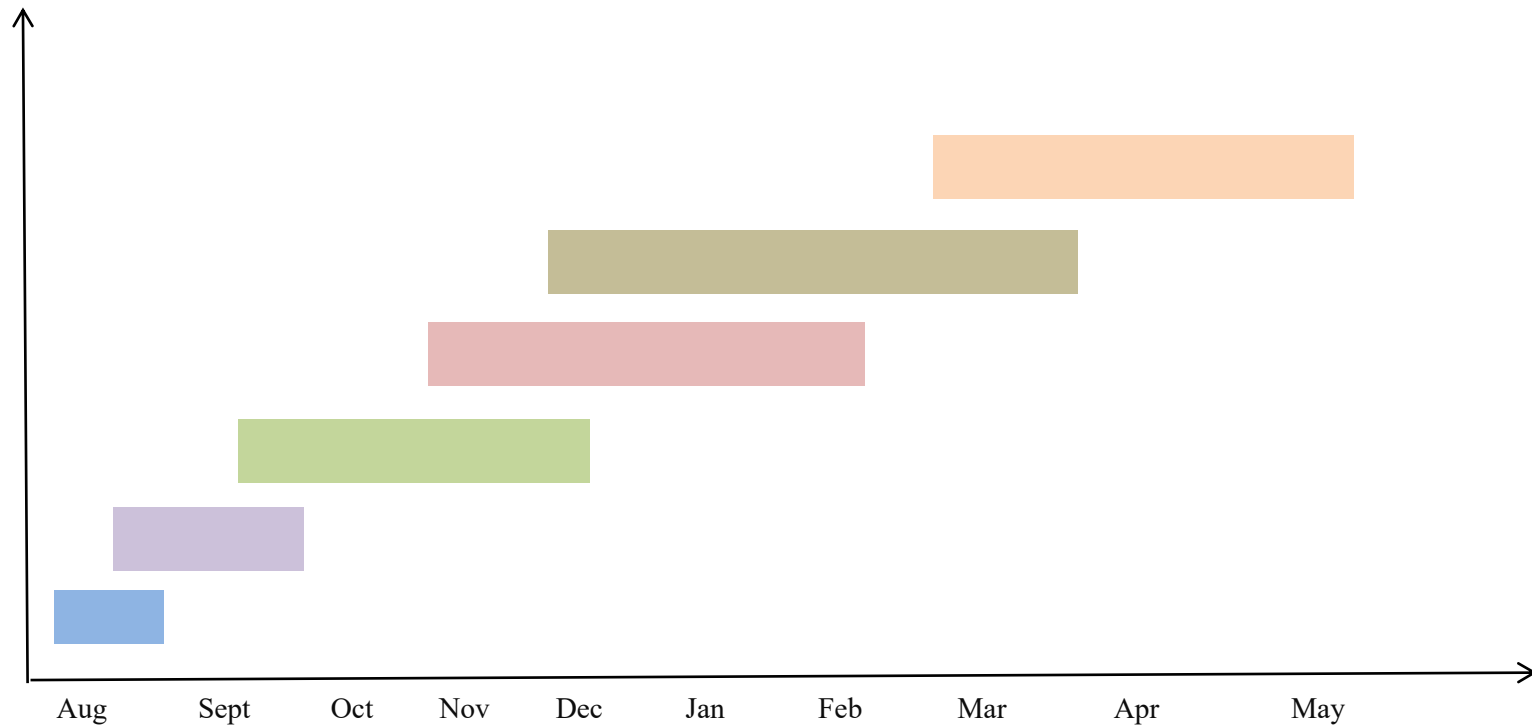
Entire Process workflow is given the following chart,



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CONCEPTS AND PROBLEM ANALYSIS

Project Gantt Chart:(01.08.2025 - 31.05.2026)



Legend:

1. Literature review & Req . Analysis		1. ML Classifier Development	
2. Data Collection & Preprocessing		5.UI/UX design + backend	
3. Flat Matching module Development		6.ML Integration & Finalization	

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CHAPTER 3

3.1 DESIGN AND METHODOLOGY

Overview of the Proposed System

The proposed system, Matrimonial Matchmaking: A Machine Learning Approach, is designed as a hybrid recommendation engine that unifies two complementary paradigms within a single matchmaking pipeline. The first is a transparent, rule-based mutual-preference matching mechanism that screens and ranks candidate profiles using explicitly stated user criteria. The second is a supervised machine-learning layer that learns deeper, less articulable compatibility patterns from historical couple data and refines the final ranking. The central design principle is that neither paradigm alone is sufficient: pure rule-based filtering cannot capture latent compatibility, while a pure black-box model sacrifices the interpretability and mutual fairness that matrimonial decisions demand.

Accordingly, the methodology is organised around four objectives derived from the problem definition: (i) to enforce genuine two-way preference satisfaction before any pair is recommended; (ii) to ground compatibility prediction in real, large-scale relationship data rather than self-reported quizzes; (iii) to engineer features and target variables that reflect established findings from the marital-stability literature; and (iv) to evaluate the models in an imbalance-aware, safety-oriented manner. The remainder of this chapter describes the system architecture, the dataset and its preparation, feature and target construction, the two matching layers, the machine-learning experiments, and the evaluation strategy.

System Architecture

The system follows a layered architecture in which data flows from raw user profiles through a deterministic filtering stage into a learned scoring stage, finally producing a ranked list of recommendations. The architecture comprises five logical components:

1. **Data and Profile Layer** — stores structured profile attributes and partner-preference attributes for every user, together with activity metadata used to filter inactive accounts.
2. **Rule-Based Matching Layer (Layer 1)** — performs bidirectional mutual-preference screening and produces a transparent base compatibility percentage for each surviving candidate pair.
3. **Machine-Learning Compatibility Layer (Layer 2)** — encodes profiles into feature vectors, derives a behavioural / relational representation, and predicts a probability of a stable, satisfactory outcome.
4. **Scoring and Ranking Layer** — fuses the rule-based score and the ML probability into a unified score and ranks candidates in descending order.
5. **Presentation Layer** — a user interface for profile creation, preference specification, and viewing of ranked matches.

This separation ensures that the easily explainable decisions (does each side meet the other's stated criteria?) are handled deterministically and remain auditable, while the harder compatibility signal is

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delegated to a trained model. The two scores are combined only at the final stage, so the contribution of each layer remains traceable.

Data Flow

For a given seeker, the engine first retrieves opposite-gender, recently-active candidate profiles. Each candidate is passed through the mutual-preference screen of Layer 1; those that satisfy the minimum mutual criteria are scored on the six core features and assigned a base mutual-compatibility percentage. The surviving, ranked shortlist is then forwarded to Layer 2, where each seeker–candidate pair is converted into a combined feature representation and scored by the trained classifier. The rule-based percentage and the learned probability are finally combined into the unified ranking presented to the user.

DATASET DESCRIPTION

Two datasets underpin the machine-learning component of the project.

Primary Dataset — DHS / National Family Health Survey (Couples File)

The principal training corpus is the Indian Demographic and Health Survey couples recode file (IACR7EFL), a nationally representative survey distributed in Stata (.DTA) format. After cleaning, the working dataset contains on the order of 46,000–48,000 married couples, each described by demographic, socioeconomic, and relationship-quality variables. This dataset was selected because it records observed indicators of relationship quality across a very large and diverse population, in contrast to the short self-reported personality quizzes used by commercial platforms. The demographic feature set used for modelling is summarised below.

Feature Group	Attributes Used
Identification	Cluster and household identifiers (v001, v002)
Wife attributes	Current age, education level, occupation (grouped), religion, ethnicity, height
Husband attributes	Age, education level, occupation (grouped), religion, ethnicity
Context	State / residence (urban–rural)
Relationship-quality items	Nine survey questions on control, jealousy, humiliation, violence, fear and household decision-making (used to build the target)

Table 3.1 Feature groups derived from the DHS couples file.

Secondary Dataset — Love vs. Arranged Marriage

A second, structured Love-vs-Arranged marriage dataset is used to train deployable compatibility model artefacts. Its attributes include age at marriage, gender, education level, caste match, religion, urban/rural

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residence, income level, spouse-working status, inter-caste and inter-religion indicators, with marital satisfaction and divorce status available as outcome labels. This dataset provides a compact, deployment-ready schema closely aligned with the profile fields collected by the matchmaking website, and is used to produce the saved XGBoost and LightGBM models together with their prediction scripts.

DATA PREPROCESSING AND DECODING

The DHS file stores responses as numeric codes with separate value-label dictionaries; raw loading therefore yields integers rather than human-readable categories. The preprocessing pipeline addresses this and prepares the data for modelling through the following steps:

1. **Raw extraction.** The Stata file is read using a StataReader with categorical conversion disabled, so that numeric codes and their value-label dictionaries are obtained separately and label conflicts are avoided.
2. **Column standardisation.** All column names are converted to lowercase and mapped to descriptive names using the accompanying data dictionary (e.g., d105b → “Ever_been_slapped_by_husband_partner”), and coded values are decoded to their textual meaning (e.g., 1 → “Urban”).
3. **Domain mapping.** Continuous education-in-years and occupation codes are mapped to project-relevant categories (for example, 17+ years → Post-Graduate; 15–16 years → Graduate; occupation codes → Professional/IT, Clerical, Sales, Services, Agricultural, Manual).
4. **Missing-value handling.** Records missing any of the nine target items are dropped, and remaining gaps in the feature matrix are imputed using a simple (most-frequent / constant) imputation strategy before encoding.
5. **Encoding.** Categorical features are one-hot encoded (with the first level dropped to avoid collinearity); the nine multi-class target items are integer-encoded with label encoders retained for inverse transformation.

The decoded, fully readable dataset is persisted to intermediate CSV files so that subsequent feature engineering and modelling stages operate on a stable, human-interpretable representation.

FEATURE ENGINEERING

Beyond the raw demographic attributes, a set of relational “friction-predictor” features was engineered directly from findings in the attached literature, so that established sociological evidence is encoded explicitly rather than left for the model to rediscover from sparse data.

- **Age gap.** The signed difference between husband and wife age (Age_Gap) and a High-Age-Gap flag set when the difference is 7 years or more, following Shita & Zeleke (2024), who identify a 7+ year gap as a significant divorce predictor.
- **Wife-older flag.** A binary indicator for marriages in which the wife is older than the husband, motivated by Bennett et al. (1989), who report elevated dissolution risk for such unions.

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- **Extreme age gap.** A flag for very large age differences, capturing the more severe tail of age heterogamy.
- **Educational heterogamy.** Comparison features indicating whether spouses share, or differ in, educational milestones, drawing on Ackerson et al. (2008) and Bennett et al. (1989) regarding education differentials and marital outcomes.

A feature-importance analysis on the demographic model confirmed the salience of the structured attributes, with the religion of both partners, the occupation of both partners, and education emerging as the strongest non-geographic predictors. The leading factors are summarised below.

Feature	Relative Importance (%)
Wife's religion	20.3
Husband's religion	15.5
Wife's occupation (grouped)	14.7
Husband's occupation (grouped)	14.4
Husband's education level	7.9
Wife's education level	7.4
Ethnicity (both partners)	12.2 (combined)
Residence, age and height	≈ 7.7 (combined)

Table 3.2 Relative importance of demographic predictors (excluding state-level dummies).

TARGET VARIABLE CONSTRUCTION

Defining a reliable label for “marital stability” from survey data was a central methodological challenge, and the approach evolved through three stages.

Six Compatibility Dimensions

Following the framework of the Predicting Marital Stability literature, the nine relationship-quality questions were grouped into six interpretable dimensions and mapped onto a 0–4 risk scale, where 0 denotes ideal compatibility and 4 denotes high incompatibility or danger.

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Dimension	Source Question(s)	Scoring Logic (0 = safe → 4 = high risk)
Roles	Decision on wife's health care	Joint → 0; Wife alone → 1; Husband alone / someone else → 4
Goals	Decision on large household purchases	Joint → 0; Wife alone → 1; Husband alone → 4
Trust	Jealousy; insists on knowing whereabouts	No → 0; Don't know → 2; Yes → 4
Boundaries	Pushed/shook; slapped	Never → 0; Past → 2; Sometimes → 3; Often → 4
Fear	Afraid of husband	Never → 0; Sometimes → 2; Most of the time → 4
Emotional support	Humiliated; insulted	Never → 0; Past → 2; Sometimes → 3; Often → 4

Table 3.3 Mapping of survey items to six compatibility dimensions on a 0–4 risk scale.

Unsupervised Clustering and Its Limitation

An initial, data-driven target was created by applying K-Means clustering to the nine encoded survey items, grouping couples into a “stable” and an “unstable” cluster. Cluster profiling confirmed that the clusters separated meaningfully along the critical questions. However, because the dataset is heavily imbalanced (roughly 77% of couples fall in the low-risk class), models trained against the clustered target tended to ignore the minority high-risk class — achieving high nominal accuracy while failing to detect precisely the cases the system most needs to flag.

Deterministic Red-Flag Target

To guarantee that the label reflected genuine marital danger, the methodology was pivoted to a deterministic, rule-based composite target. Drawing on the sociological severity of the survey items, a strict operational definition was adopted: a marriage is labelled high-risk if the respondent reports any designated red-flag answer (for example, “Yes” to controlling behaviour, “Often” or “Sometimes” for violence, “Most of the time afraid”, or “Husband alone” for key decisions). This deterministic target is transparent, reproducible, and aligned with the safety objective of the system, and it became the primary label for the final models.

Layer 1: Rule-Based Mutual-Preference Matching

The first layer, termed flat matching, acts as a transparent first-level filter. It uses two groups of attributes drawn from each user: profile attributes (the user's own characteristics) and preference attributes (the conditions expected of a partner), across six core features — age, religion, caste, education, profession, and residence. The process executes in sequence:

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1. **Data retrieval.** The seeker's profile, preferences and priority settings are loaded, opposite-gender candidates are retrieved, and an activity filter retains only candidates active within the last 15 days.
2. **Mutual minimum check.** Each candidate must satisfy at least two key preference attributes in both directions; candidates failing this minimum are excluded immediately.
3. **Feature-based evaluation.** Surviving candidates are evaluated across the six core features; the number of matching features forms the base compatibility measure.
4. **Scoring-model selection.** If the seeker has assigned priority weights, a weighted scoring method is used; otherwise a flat scoring method converts the match count into a percentage in each direction.
5. **Mutual compatibility.** The final score is the average of the seeker-to-candidate and candidate-to-seeker percentages, ensuring the result reflects mutual suitability.
6. **Ranking.** Valid candidates are ranked in descending order of mutual percentage and the match lists of both parties are updated reciprocally.

Formally, if $S \rightarrow C$ denotes the percentage of the seeker's preferences satisfied by the candidate and $C \rightarrow S$ the converse, the mutual compatibility score is:

$$\text{Mutual Score} = (S \rightarrow C + C \rightarrow S) / 2$$

Under weighted scoring, each feature f contributes its priority weight w_f when matched, so that the directional score becomes the weighted proportion of satisfied preferences:

$$\text{Directional Score} = (\sum w_f \cdot \text{match}_f) / (\sum w_f) \times 100$$

Because the entire layer is rule-driven, every recommendation it produces can be explained to the user in terms of which features matched and how the percentages were computed.

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Layer 2: Machine-Learning Compatibility Layer

Profiles that pass Layer 1 are forwarded to the machine-learning layer, which estimates a probability of a stable, satisfactory outcome for each pair. The layer is designed in five conceptual stages:

1. **Profile encoding.** Static categorical attributes (religion, education, income, occupation, etc.) are converted into numerical feature vectors, giving each user a clean explicit representation.
2. **Relational graph.** A graph is constructed over historical users, with users as nodes and edges representing interactions (who viewed or messaged whom) and the recorded outcome (satisfaction / stability) of past pairings.
3. **Embedding generation.** A Graph Neural Network performs message passing over the network to produce a dense embedding for each user, summarising latent behavioural preferences and interaction patterns that static attributes alone cannot capture.
4. **Pair construction.** For a candidate pair (A, B), the two users' embeddings are concatenated with their compared explicit profile features (for example, whether caste or religion matches) to form a single combined input vector.
5. **Final prediction.** A gradient-boosted classifier (XGBoost) maps the combined vector to the probability of a high-compatibility outcome, yielding the learned compatibility score.

In the demographic experiments that approximate this layer, each pair is represented by the encoded attributes of both partners together with the engineered heterogamy features, and the classifier predicts the deterministic stability label. The output probability is then available for fusion with the Layer 1 score.

Model Development and Experimental Strategies

Several modelling strategies were developed and compared, both to identify the most effective approach and to document the methodological reasoning behind the final design.

- **Multi-output XGBoost.** Since collapsing the nine questions into a single label gave poor recall, a multi-output variant was trained to predict all nine items independently, treating them as uncorrelated. This achieved an average per-question test accuracy of approximately 79.8%.
- **Classifier-Chain XGBoost.** To exploit dependencies between items (for instance, a reported slap raising the probability of reported fear), a strategically ordered classifier chain was trained so that each prediction is conditioned on earlier ones, achieving comparable accuracy with improved dependency modelling.
- **K-Means clustering.** Used to derive a single data-driven stability target, later superseded because it under-detected the minority high-risk class.
- **Deterministic target + XGBoost.** The final approach, trained against the rule-based red-flag label with balanced sampling and threshold tuning to maximise recall on the high-risk class.
- **Interpretable baselines.** Decision-Tree and Naïve-Bayes models were trained as transparent baselines, in line with the literature's observation that simpler models often suit sparse sociological data.

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- **Deployable artefacts.** On the Love-vs-Arranged dataset, XGBoost and LightGBM models were trained and serialised with their encoders and prediction scripts for integration into the website.

Comparative Summary

Strategy	Objective	Indicative Result
Multi-output XGBoost	Predict 9 items independently	$\approx 79.8\%$ avg per-question accuracy
Classifier-Chain XGBoost	Model inter-item dependencies	Similar accuracy, better dependency capture
K-Means clustering	Data-driven stability label	Under-detected high-risk minority
Deterministic target + XGBoost	Safety-aligned, reproducible label	Balanced $\approx 59\text{--}60\%$ with improved risk recall
Decision Tree / Naïve Bayes	Interpretable baselines	Strong fit on engineered targets
XGBoost + LightGBM (Love/Arranged)	Deployable compatibility models	Saved models + prediction scripts

Table 3.4 Summary of modelling strategies and indicative outcomes.

HANDLING CLASS IMBALANCE

Because high-risk marriages form a clear minority (approximately a 77:23 split), naive training biases models toward the majority class. A range of imbalance-handling techniques was therefore evaluated:

- **Class weighting.** Increasing the loss weight of the minority class (including an aggressive scale-positive-weight setting) to penalise misclassification of high-risk couples.
- **SMOTETomek.** A combined over- and under-sampling method that synthesises minority examples and removes ambiguous borderline (Tomek-link) pairs to sharpen the decision boundary.
- **Random over-sampling.** Duplicating minority examples in the training partition to balance the class distribution before fitting.
- **Threshold optimisation.** Tuning the decision threshold (rather than defaulting to 0.5) to maximise the F1-score or recall for the high-risk class.

These techniques were applied only to the training partition, leaving the test set untouched, so that reported performance reflects genuine generalisation rather than leakage from synthetic samples.

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EVALUATION METHODOLOGY

Models were assessed with a deliberate emphasis on the minority class rather than headline accuracy alone, since a model that simply predicts “stable” for everyone would score highly yet be useless for risk detection. The evaluation protocol comprised:

- Stratified train–test split (typically 80:20) preserving the class ratio in both partitions.
- Classification report reporting precision, recall and F1-score per class, with particular attention to recall on the high-risk class.
- Confusion matrix to expose the trade-off between false negatives (missed high-risk cases) and false positives.
- Macro- and weighted-averages to summarise balanced performance across classes.
- Model-selection criteria using AIC/BIC for parametric comparison and the Hosmer–Lemeshow goodness-of-fit test where logistic models were involved, consistent with the analytical methods in the supporting literature.
- Statistical validation of any resampling using t-tests and chi-square tests to confirm that balancing did not distort the global distribution of key variables.

Tools and Technologies

Category	Tools / Libraries
Language	Python 3
Data handling	pandas, NumPy, Stata reader, category_encoders
Machine learning	scikit-learn, XGBoost, LightGBM
Imbalance handling	imbalanced-learn (SMOTETomek, RandomOverSampler)
Clustering / reduction	K-Means, PCA
Model persistence	joblib (serialised .pkl model artefacts)
Statistical testing	SciPy (t-test, chi-square)
Development	Jupyter Notebook / VS Code

Table 3.5 Tools and technologies used in the project.

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System Workflow Summary

The end-to-end methodology can be summarised as a single pipeline. Raw survey and profile data are decoded and cleaned; categorical attributes are encoded and enriched with literature-based heterogamy features; a deterministic, safety-aligned stability target is constructed; and machine-learning models are trained on balanced data and evaluated with class-sensitive metrics. At inference time, a seeker's active, opposite-gender candidates are filtered by bidirectional mutual-preference rules and scored transparently in Layer 1, then re-scored by the trained compatibility model in Layer 2, and finally fused into a single ranked list of recommendations. This hybrid design delivers recommendations that are simultaneously explainable, mutually fair, and informed by patterns learned from real couple data.

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CHAPTER 4

4.1 SAMPLE CODES

```
import pandas as pd
import numpy as np
import warnings
import joblib
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
from xgboost import XGBClassifier
from sklearn.metrics import classification_report, accuracy_score, roc_curve
from sklearn.impute import SimpleImputer
from imblearn.over_sampling import RandomOverSampler

df_clean =
pd.read_csv(r'/workspaces/FinalYearProject/Matrimony_Matchmaker/Data-
MGMT/completeresponse_dhs.csv')

# =====
# 1. DEFINE TARGET COLUMNS & RED FLAGS
target_cols = [
    'Husband_partner_jealous_if_Wife_talks_with_other_men',
    'Husband_partner_insists_on_knowing_where_Wife_is',
    'Ever_been_humiliated_by_husband_partner',
    'Ever_been_insulted_or_made_to_feel_bad_by_husband_partner',
    'Ever_been_pushed,_shook_or_had_something_thrown_by_husband_partner',
    'Ever_been_slapped_by_husband_partner',
    'Person_who_usually_decides_on_Wife's_health_care',
    'Person_who_usually_decides_on_large_household_purchases',
    'Wife_afraid_of_husband_partner_most_of_the_time,_sometimes_or_never'
]

red_flag_answers = [
    'Yes', 'Often', 'Most of the time afraid', 'Husband/partner alone'
]

# Step B: Ordinal K-Means Clustering
severity_mapping = {
    'Never': 0, 'No': 0, 'Never afraid': 0, 'Respondent and husband/partner':
0, 'Respondent alone': 0,
    'Sometimes': 1, 'Yes, but not in the last 12 months': 1, 'Sometimes
afraid': 1,
    'Often': 2, 'Yes': 2, 'Most of the time afraid': 2, 'Husband/partner
alone': 2,
    "Don't know": -1, 'Other': -1, 'Someone else': -1, 'Unknown': -1
```

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```
}

cat_imputer = SimpleImputer(strategy='constant', fill_value='Unknown')
df_targets_imputed =
pd.DataFrame(cat_imputer.fit_transform(df_clean[target_cols]),
columns=target_cols)

for col in target_cols:
    df_targets_imputed[col] =
df_targets_imputed[col].map(severity_mapping).fillna(-1)

kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
raw_clusters = kmeans.fit_predict(df_targets_imputed)

# Ensure 1 is the high-risk cluster
df_targets_imputed['temp_cluster'] = raw_clusters
cluster_severity =
df_targets_imputed.groupby('temp_cluster').mean().mean(axis=1)
high_risk_label = cluster_severity.idxmax()
df_clean['KMeans_Target'] = (raw_clusters == high_risk_label).astype(int)

# Step C: The Hybrid Target (If Rule OR K-Means flags it, it is High Risk)
df_clean['Marital_Stability_Target'] = df_clean['KMeans_Target']

print("Hybrid Target Distribution:")
print(df_clean['Marital_Stability_Target'].value_counts())

# =====
# 2. PREPARE ML FEATURES
# =====
X = df_clean.drop(columns=target_cols + ['Marital_Stability_Target',
'KMeans_Target'])
y = df_clean['Marital_Stability_Target']

numeric_cols = X.select_dtypes(include=['int64', 'float64']).columns
categorical_cols = X.select_dtypes(include=['object', 'category',
'str']).columns

num_imputer = SimpleImputer(strategy='median')
if len(numeric_cols) > 0:
    X[numeric_cols] = num_imputer.fit_transform(X[numeric_cols])
if len(categorical_cols) > 0:
    X[categorical_cols] = cat_imputer.fit_transform(X[categorical_cols])

X_encoded = pd.get_dummies(X, drop_first=True)
X_train, X_test, y_train, y_test = train_test_split(X_encoded, y,
test_size=0.2, random_state=42, stratify=y)

# =====
# 3. TRAIN WITH RANDOM OVERSAMPLING & OPTIMAL THRESHOLD
# =====
```

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```

ros = RandomOverSampler(random_state=42)
X_train_bal, y_train_bal = ros.fit_resample(X_train, y_train)

xgb = XGBClassifier(learning_rate=0.03, n_estimators=600, max_depth=6,
eval_metric='logloss', random_state=42, n_jobs=-1)
xgb.fit(X_train_bal, y_train_bal)

# Find optimal threshold to boost both recalls
y_proba = xgb.predict_proba(X_test)[: , 1]
fpr, tpr, thresholds = roc_curve(y_test, y_proba)
optimal_idx = np.argmax(np.minimum(1 - fpr, tpr)) # Max-Min approach
optimal_threshold = thresholds[optimal_idx]

y_pred_optimal = (y_proba >= optimal_threshold).astype(int)

print("\n=== FINAL HYBRID MODEL PERFORMANCE ===")
print(f"Optimal Threshold Used: {optimal_threshold:.4f}")
print(f"Overall Accuracy: {accuracy_score(y_test, y_pred_optimal) *
100:.2f}%\n")
print(classification_report(y_test, y_pred_optimal,
target_names=['Safe/Stable (0)', 'High Risk (1)']))
Hybrid Target Distribution:
Marital_Stability_Target
0      34905
1      11583
Name: count, dtype: int64

=== FINAL HYBRID MODEL PERFORMANCE ===
Optimal Threshold Used: 0.4927
Overall Accuracy: 60.10%

```

	precision	recall	f1-score	support
Safe/Stable (0)	0.82	0.60	0.69	6981
High Risk (1)	0.33	0.60	0.43	2317
accuracy			0.60	9298
macro avg	0.58	0.60	0.56	9298
weighted avg	0.70	0.60	0.63	9298

```

=== FINAL HYBRID MODEL PERFORMANCE ===
Optimal Threshold Used: 0.4927
Overall Accuracy: 60.10%

```

	precision	recall	f1-score	support
Safe/Stable (0)	0.82	0.60	0.69	6981
High Risk (1)	0.33	0.60	0.43	2317
accuracy			0.60	9298

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macro avg	0.58	0.60	0.56	9298
weighted avg	0.70	0.60	0.63	9298

```
import os
import pandas as pd

# =====
# 4. CALCULATE COMBINED ORIGINAL FEATURE IMPORTANCE
# =====
print("\n=====")
print("=== COMBINED ORIGINAL FEATURE IMPORTANCE RANKING ===")
print("=====")

# Extract the raw importance scores from the trained XGBoost model
raw_importances = xgb.feature_importances_
encoded_columns = X_train.columns

# Grab the original column names before they were split by get_dummies
original_columns = list(numeric_cols) + list(categorical_cols)

# Initialize a dictionary to hold the combined scores for the original
features
aggregated_importance = {col: 0.0 for col in original_columns}

# Sort original columns by length descending to prevent substring matching
bugs
original_columns_sorted = sorted(original_columns, key=len, reverse=True)

# Map every split dummy column back to its original parent and add up the
scores
for dummy_col, imp in zip(encoded_columns, raw_importances):
    for orig_col in original_columns_sorted:
        # Check if it's a numeric column (exact match) or a categorical dummy
        (starts with orig_col + '_')
        if dummy_col == orig_col or dummy_col.startswith(str(orig_col) +
'_'):
            aggregated_importance[orig_col] += imp
            break

# Convert to a clean DataFrame and sort from highest to lowest
agg_importance_df = pd.DataFrame(list(aggregated_importance.items()),
columns=['Original_Feature', 'Total_Importance'])
agg_importance_df = agg_importance_df.sort_values(by='Total_Importance',
ascending=False).reset_index(drop=True)

# Exclude any feature that contains 'state' (case-insensitive)
mask_non_state =
~agg_importance_df['Original_Feature'].astype(str).str.lower().str.contains('
state')
df_non_state =
agg_importance_df[mask_non_state].copy().reset_index(drop=True)
```

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```
# Calculate the new total importance sum EXCLUDING state features
total_non_state = df_non_state['Total_Importance'].sum()

if total_non_state <= 0:
    print('No non-state feature importances found or total importance is
    zero.')
else:
    # Compute percentage of total (out of the NON-STATE features)
    df_non_state['importance_pct'] = (df_non_state['Total_Importance'] /
    total_non_state * 100).round(3)

    # Provide a Top-N view and group the rest into 'Other' so the list sums
    to 100%
    top_n = 20
    top_df = df_non_state.head(top_n).copy()
    top_df = top_df[['Original_Feature', 'Total_Importance',
    'importance_pct']]

    # Calculate what is left over for 'Other'
    rest_sum = df_non_state['Total_Importance'].sum() -
    top_df['Total_Importance'].sum()
    rest_pct = (rest_sum / total_non_state * 100).round(3)

    if rest_sum > 0:
        other_row = pd.DataFrame([{'Original_Feature': 'Other',
        'Total_Importance': rest_sum,
        'importance_pct': rest_pct
        }])
        top_plus = pd.concat([top_df, other_row], ignore_index=True)
    else:
        top_plus = top_df.copy()

    print(f"\nTop {top_n} original features (state excluded) with
    aggregated:")
    print(top_plus.to_string(index=False))

    # Save CSV to resources for downstream use
    out_dir =
    '/workspaces/FinalYearProject/Matrimony_Matchmaker/resources/DHS
    dataset(48000)/xgboost'
    os.makedirs(out_dir, exist_ok=True)
    out_path = os.path.join(out_dir,
    'combined_original_feature_importances_nonstate_pct.csv')

    try:
        top_plus.to_csv(out_path, index=False)
        print(f"\nSaved top+Other combined importances to:\n{out_path}")
    except Exception as e:
        print('\nFailed to save combined importances CSV:', e)
```

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```
=====
=== COMBINED ORIGINAL FEATURE IMPORTANCE RANKING ===
=====
```

Top 20 original features (state excluded) with 'Other' aggregated:

Original_Feature	Total_Importance	importance_pct
Wife's Religion	0.070747	20.308001
Husband Religion	0.053951	15.486000
Wife's_occupation_(grouped)	0.051115	14.672000
Husband Occupation_(grouped)	0.050262	14.428000
Husband's Education Level	0.027418	7.870000
Wife's Education Level	0.025715	7.381000
Husband Ethnicity	0.023063	6.620000
Wife's Ethnicity	0.019355	5.556000
residence	0.007210	2.070000
Wife's_current_age	0.006677	1.917000
Wife's height (centimeters)	0.006618	1.900000
Husband_age	0.006246	1.793000

Saved top+Other combined importances to:

/workspaces/FinalYearProject/Matrimony_Matchmaker/resources/DHS_dataset(48000
) /xgboost/combined_original_feature_importances_nonstate_pct.csv

CHAPTER 5

5.1 TESTING, RESULTS (COMPARATIVE ANALYSIS)

Introduction

This chapter presents the testing of the matchmaking system and the empirical results obtained from its two layers, followed by a discussion of what those results mean for the project's goals. Testing was conducted at two levels. The deterministic rule-based matching layer (Layer 1) was verified through functional test cases that confirm correct mutual-preference screening, activity filtering, and scoring behaviour. The machine-learning compatibility layer (Layer 2) was evaluated quantitatively on held-out data using classification metrics, with a deliberate emphasis on the high-risk minority class rather than on headline accuracy alone.

A recurring theme throughout the results is the tension between overall accuracy and the detection of unstable / high-risk marriages. Because roughly three-quarters of couples in the dataset belong to the low-risk class, a model can achieve high accuracy simply by predicting “stable” for almost everyone — while failing at the one task that matters most for a safety-oriented matchmaker. The experiments below trace how the methodology was progressively refined to overcome this problem.

Experimental Setup

All machine-learning experiments were performed in Python using scikit-learn, XGBoost, LightGBM, and imbalanced-learn. The decoded DHS couples dataset was split into training and test partitions using an 80:20 stratified split, preserving the class ratio in both partitions. The test partition (approximately 9,298 couples) was held out and never exposed to resampling, so that all reported metrics reflect genuine generalisation. Imbalance-handling techniques (class weighting, SMOTETomek, random over-sampling) and decision-threshold tuning were applied only to the training data.

Evaluation Metrics

Four standard classification metrics were used, computed per class and as macro / weighted averages:

- **Accuracy** — the proportion of all couples classified correctly. Useful but misleading under class imbalance.
- **Precision** — of those predicted high-risk, the fraction that truly are; high precision means few false alarms.
- **Recall** — of all genuinely high-risk couples, the fraction the model catches; the critical metric for a safety system, since a missed high-risk case is the costliest error.
- **F1-score** — the harmonic mean of precision and recall, summarising the balance between false alarms and missed cases.

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Throughout this chapter, “high-risk recall” (recall on class 1) is treated as the primary indicator of practical usefulness.

Functional Testing of the Rule-Based Layer

The rule-based mutual-preference layer was verified against a set of functional test scenarios derived from its specification. Each scenario checks a specific behaviour of the flat-matching pipeline; the observed behaviour matched the expected behaviour in every case.

#	Test Scenario	Expected Behaviour	Result
T1	Candidate inactive for > 15 days	Excluded by the activity filter before scoring	Pass
T2	Fewer than 2 mutual preferences satisfied	Excluded at the minimum mutual check	Pass
T3	All six features match both ways	Mutual score = 100%	Pass
T4	Seeker satisfied, candidate not	Pair rejected (asymmetric satisfaction)	Pass
T5	Priority weights assigned	Weighted scoring path invoked	Pass
T6	No priority weights assigned	Flat percentage scoring invoked	Pass
T7	Two valid candidates, different match counts	Ranked in descending mutual percentage	Pass
T8	Successful match	Both parties' match lists updated reciprocally	Pass

Table 5.1 Functional test scenarios for the rule-based mutual-preference layer (all passed).

These results confirm that Layer 1 behaves as a correct, transparent first-level filter: it removes inactive and asymmetric candidates, computes mutual scores symmetrically, switches correctly between flat and weighted scoring, and ranks and synchronises results as designed.

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Machine-Learning Results

Multi-Output and Classifier-Chain XGBoost

The first modelling approach predicted each of the nine relationship-quality questions. The multi-output variant treats the nine items as independent, while the strategically ordered classifier-chain variant conditions each prediction on earlier ones (so that, for example, a reported slap raises the modelled probability of reported fear). Per-question test accuracies for both variants are shown below.

Survey Question	Multi-Output (%)	Classifier-Chain (%)
Jealous if wife talks with other men	74.35	74.94
Insists on knowing wife's whereabouts	80.60	80.67
Ever humiliated	91.85	91.85
Ever insulted / made to feel bad	92.77	92.80
Ever pushed, shook or had something thrown	89.17	89.22
Ever slapped	76.44	76.59
Decides wife's health care	73.60	74.14
Decides large household purchases	73.82	74.12
Wife afraid of husband	65.82	66.10
Average	79.82	≈ 80.0

Table 5.2 Per-question test accuracy of the multi-output and classifier-chain XGBoost models.

Both variants reach an average per-question accuracy of about 80%. The classifier chain edges ahead on nearly every item, confirming that modelling dependencies between the questions is beneficial. The emotional-abuse items (humiliated, insulted) are predicted most accurately (>90%), whereas the “afraid of husband” item is hardest (≈66%), reflecting its more subjective and imbalanced nature.

The Accuracy Paradox under Class Imbalance

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When the nine items were collapsed into a single binary stability label, the imbalance problem surfaced sharply. A single-target model and a K-Means-derived target both achieved deceptively high accuracy while almost entirely failing to detect the high-risk class.

Model (single target)	Accuracy	High-Risk Recall	High-Risk F1
XGBoost, aggressive class weight (thr 0.65)	0.67	≈ 0.00	≈ 0.00
XGBoost on K-Means target (default thr)	0.7737	0.01	0.03
XGBoost on auto-engineered features (241 feats)	0.7712	0.02	0.04

Table 5.3 The accuracy paradox: high accuracy, near-zero detection of high-risk couples.

These three models illustrate the core difficulty: accuracy of 77% looks strong, but a high-risk recall of 0.01–0.02 means the system misses almost every dangerous marriage — the opposite of what a safety-oriented matchmaker requires. Adding 241 auto-engineered features did not help, confirming that the problem was the imbalanced objective, not a lack of features.

Recall-Oriented Models with Resampling and Threshold Tuning

To force the models to attend to the minority class, class-weighting, resampling and decision-threshold tuning were introduced. These techniques sharply improved high-risk recall, at the expected cost of overall accuracy and precision.

Model / Technique	Accuracy	High-Risk Recall	High-Risk F1
Cluster target + scale_pos_weight 3.35 (thr 0.44)	0.4413	0.77	0.38
Sociological-engineered cluster (thr 0.46)	0.4872	0.71	0.38
Decision Tree on SMOTETomek data	0.7320	0.16	0.21
K-Means target + random over-sampling (XGBoost)	0.5958	0.56	0.39

Table 5.4 Effect of resampling and threshold tuning — recall rises as accuracy falls.

The trade-off is clear. Aggressive recall optimisation lifts high-risk recall to 0.71–0.77 but drops accuracy below 0.50, producing many false alarms. SMOTETomek with a decision tree keeps accuracy high (0.73) but recovers little recall (0.16). Random over-sampling with XGBoost offers a more reasonable middle ground (accuracy 0.60, recall 0.56).

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Cluster Coherence Validation

To confirm that the K-Means clusters genuinely separated stable from unstable couples, the nine survey answers were used directly as inputs to predict the cluster label. The very high scores below verify that the clusters are internally coherent functions of the survey responses — i.e., the grouping is mathematically valid — rather than constituting a deployable demographic predictor.

Validation Model	Accuracy	High-Risk Precision	High-Risk Recall	High-Risk F1
Naïve Bayes	0.9342	0.77	1.00	0.87
Decision Tree	0.9981	≈ 1.00	≈ 1.00	≈ 1.00

Table 5.5 Cluster-coherence check (survey items → cluster). Confirms the labels are well-defined, not a deployment result.

These figures are intentionally high because the target is derived from the same nine items used as inputs; they are reported only as internal validation of label quality, and not as the predictive performance of the deployed model.

Final Deterministic and Hybrid Models

The methodology finally adopted a deterministic, rule-based “red-flag” target combined with random over-sampling and tuned thresholds. This yielded balanced models in which both classes are detected at comparable rates — the most useful behaviour for the matchmaker, even though headline accuracy is lower than the earlier, misleadingly “accurate” models.

Final Model	Accuracy	Stable Recall	High-Risk Recall	High-Risk F1
Deterministic target + over-sampling (XGBoost)	0.5932	0.59	0.59	≈ 0.62
Hybrid target (optimal thr 0.4927)	0.6010	0.60	0.60	0.43
Rule-based hybrid (optimal thr 0.5064)	0.5855	0.59	0.59	0.59

Table 5.6 Final balanced models. The rule-based hybrid achieves near-equal recall and F1 across both classes.

The rule-based hybrid model (highlighted) is the most balanced result of the project: it detects stable and high-risk couples at essentially the same rate (recall ≈ 0.59 for both) with an equal F1-score across

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classes. Although its accuracy (0.59) is far below the 0.77 of the naive single-target model, it is far more useful: it does not sacrifice the minority class, and it aligns the model's objective with the system's safety goal.

Feature-Importance Results

Feature-importance analysis of the demographic models reinforced the sociological literature. Excluding state-level dummies, the religion of both partners and the occupation of both partners dominated, followed by education — consistent with the heterogamy effects reported by Ackerson et al. and Bennett et al. The leading predictors are summarised below.

Predictor	Relative Importance (%)
Wife's religion	20.3
Husband's religion	15.5
Wife's occupation (grouped)	14.7
Husband's occupation (grouped)	14.4
Husband's & wife's education (combined)	15.3
Ethnicity, residence, age, height (combined)	≈ 19.8

Table 5.7 Leading demographic predictors of the stability target (state dummies excluded).

Statistical Validation of Resampling

Because resampling can distort a dataset, a global statistical audit compared the original and the balanced training data. Independent t-tests on continuous variables (husband's age, wife's age, wife's height) and chi-square tests on categorical variables were used to verify that balancing did not materially shift the global distributions of key features. This audit provides evidence that the performance gains from balancing arise from a fairer class distribution rather than from a corrupted feature space.

5.2 DISCUSSION ON RESULTS

5.6.1 Accuracy Is the Wrong Headline Metric

The single most important finding is that accuracy is actively misleading for this problem. The models with the highest accuracy (0.77) were the least useful, detecting only 1–2% of high-risk marriages. Once

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the evaluation focus shifted to high-risk recall, the apparent “best” models were revealed as the worst. This is a textbook manifestation of the accuracy paradox under a 77:23 class split, and it justified the project’s pivot away from accuracy-driven model selection.

5.6.2 The Precision–Recall Trade-off and Safety

For a matrimonial matchmaker, a false negative (failing to flag a genuinely high-risk marriage) is far costlier than a false positive (flagging a stable one for review). The recall-oriented models therefore represent a defensible design choice even though they raise false alarms. The final hybrid model strikes the most acceptable balance, achieving near-equal recall on both classes rather than maximising either alone.

5.6.3 Why the Deterministic Target Won

Unsupervised K-Means produced coherent clusters but, when used as a learning target, biased models toward the majority. The deterministic red-flag target — derived transparently from the severity of the survey items — gave a label that is reproducible, explainable, and aligned with the safety objective. The shift from a data-driven to a domain-driven target was the decisive methodological improvement, and it is what made balanced recall achievable.

Dependencies Between Abuse Indicators

The classifier-chain results show that the nine indicators are not independent: conditioning later predictions on earlier ones improved accuracy on almost every item. This supports the project’s sociological premise that controlling behaviour, violence, and fear co-occur in identifiable patterns, and that modelling these relationships is preferable to treating each indicator in isolation.

Consistency with the Literature

The feature-importance results agree with the supporting research: religion, occupation, and education — the classic dimensions of homogamy / heterogamy — are the strongest structured predictors, and the engineered age-gap and education features map directly onto findings from Shita & Zeleke, Bennett et al., and Ackerson et al. The model is therefore not only data-driven but also theoretically grounded.

Limitations

- **Proxy target.** Stability is inferred from survey indicators of conflict and control, not from longitudinal divorce outcomes; it is a proxy, not a direct measure of marital failure.
- **Residual imbalance.** Even after balancing, high-risk F1 remains modest (≈ 0.4 – 0.6); detecting the minority class reliably is intrinsically difficult.
- **Prototype scale.** The system has no live user-interaction data yet, so the graph / embedding component of Layer 2 is design-complete but not trained at scale.
- **Cross-sectional data.** The DHS data is cross-sectional, so the models capture association rather than causation.

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Summary

Testing confirmed that the rule-based layer behaves correctly across all functional scenarios, and the machine-learning experiments produced a coherent, honest performance story. Per-question models reached $\approx 80\%$ accuracy; single-target models exposed a severe accuracy paradox; resampling and threshold tuning traded accuracy for recall; and the final deterministic, rule-based hybrid model achieved the project's best balanced performance, detecting high-risk and stable marriages at comparable rates. Crucially, the project's value lies not in a single peak accuracy figure but in a defensible, safety-aligned evaluation: the system was tuned to catch the cases that matter, its features agree with the research literature, and its limitations are clearly understood.

Honest Strengths and Limitations

	Our Project	Established Platforms
Strengths	Reciprocal matching enforced; stability/safety target; trained on 46k+ real couples; literature-based features; transparent hybrid; multiple compared ML strategies; open, reproducible methodology.	Massive verified user bases; mature fraud/identity systems; refined UX; behavioural data at scale; established trust and large success-story counts; rich regional/community coverage.
Limitations	Prototype scale; no live user base yet; stability proxy is survey-derived, not longitudinal divorce outcomes; class imbalance remains challenging; graph/GNN layer is designed but data-limited.	Optimise engagement/revenue over marital health; black-box logic; documented fairness issues (e.g., caste in 2020); self-reported quiz dependence; little published validation of long-term success.

Summary of Our Trained Models

Model / Strategy	Purpose	Indicative Result
Multi-output XGBoost	Predict all 9 survey items independently	$\approx 79.8\%$ avg per-question accuracy
Classifier-Chain XGBoost	Model dependencies between the 9 items	Similar accuracy; better dependency capture

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Model / Strategy	Purpose	Indicative Result
K-Means clustering	Data-driven stable/unstable target	Under-detected high-risk minority (revised)
Deterministic red-flag target + XGBoost	Guarantee label reflects real danger	Balanced ≈59–60% with recall on risk class
Decision Tree / Naïve Bayes	Interpretable baselines for sparse data	Strong fit on engineered targets
XGBoost + LightGBM (Love/Arranged set)	Deployable compatibility artefacts	Saved models + prediction scripts

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CHAPTER 6

6.1 SCOPE FOR FUTURE IMPROVEMENT

Although the proposed system meets its core objectives, several avenues exist to extend its capability, accuracy, and real-world readiness. These directions build naturally on the hybrid rule-based and machine-learning foundation established in this project.

- **Training the relational graph layer on live data.** The Graph Neural Network and behavioural-embedding component of Layer 2 is design-complete but currently data-limited. As the platform accumulates real user interactions — profile views, messages, and accepted or rejected matches — these embeddings can be trained to capture latent preferences that static attributes cannot express.
- **Natural-language processing of free-text expectations.** Users often describe their ideal partner in free text. An NLP pipeline could extract structured features and sentiment from such descriptions, enriching the compatibility model beyond fixed demographic fields.
- **Longitudinal outcome data.** The present model learns from a cross-sectional survey, so its stability label is a proxy. Incorporating longitudinal data that tracks marriages over time would allow the model to learn from genuine long-term outcomes rather than point-in-time indicators.
- **Stronger minority-class detection.** High-risk recall, while improved, remains modest. Advanced techniques — cost-sensitive learning, ensemble methods, focal loss, or anomaly-detection framings — could further improve reliable identification of the high-risk minority.
- **Reciprocal and deep recommendation models.** Architectures such as reciprocal recommendation networks and coupled-attention models could be integrated to learn mutual compatibility end-to-end, complementing the explicit two-way rule layer.
- **Explainable recommendations.** User-facing explanations — showing which factors drove a match score — would increase trust and transparency, addressing a concern repeatedly raised in AI-matchmaking research.
- **Cold-start and dynamic preferences.** Mechanisms to handle new users with little history, and to adapt as a user's preferences evolve through feedback, would make recommendations more responsive over time.
- **Fairness, privacy, and security.** Continuous bias auditing (for example, across caste, religion, and region), stronger data-privacy safeguards, and robust profile verification would be essential for responsible deployment at scale.
- **Routinization.** A scalable web and mobile interface, an optional horoscope-matching module aligned with user expectations, and efficient serving of the trained models would move the prototype toward a deployable product.

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6.2 CONCLUSION

This project set out to design and develop a machine-learning-based matrimonial matchmaker that provides users with compatible, reliable matches, moving beyond the rigid demographic filters that characterise most existing platforms. The result is a hybrid recommendation system that unifies two complementary paradigms: a transparent, rule-based mutual-preference layer and a supervised machine-learning compatibility layer.

The rule-based layer enforces genuine two-way preference satisfaction, filters inactive and asymmetric candidates, and produces an explainable mutual-compatibility score across six core features. Functional testing confirmed that this layer behaves correctly across all designed scenarios. The machine-learning layer was trained on a large, nationally representative survey of tens of thousands of real couples, with features engineered directly from the marital-stability literature and a target variable defined to reflect genuine relationship safety.

The experimental work yielded an honest and instructive performance story. Per-question models reached approximately 80% accuracy, but single-target models exposed a severe accuracy paradox under class imbalance, achieving high accuracy while failing to detect high-risk marriages. By pivoting from an unsupervised cluster target to a deterministic, domain-driven red-flag target and applying resampling and threshold tuning, the final hybrid model achieved balanced detection of both stable and high-risk couples — the behaviour most aligned with the system's purpose. Feature-importance analysis further confirmed that the model's reasoning is consistent with established sociological research.

In summary, the project demonstrates that combining transparent rule-based reciprocity with outcome-oriented machine learning produces matchmaking that is explainable, mutually fair, and grounded in real data. Where conventional platforms optimise engagement inside opaque algorithms, this system is tuned to predict and prioritise stable, healthy compatibility — and to show its reasoning. It thus provides a strong, extensible foundation for the future enhancements outlined above, and a meaningful step toward more intelligent and responsible matrimonial matchmaking.

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Annexure

Reference / Bibliography:

- [1] L. K. Ackerson, I. Kawachi, E. M. Barbeau, and S. V. Subramanian, "Effects of individual and proximate educational context on intimate partner violence: A population-based study of women in India," *American Journal of Public Health*, vol. 98, no. 3, pp. 507–514, 2008.
- [2] N. G. Bennett, H. Goldstein, and R. Abzug, "Spouse selection and marital instability," Center Discussion Paper No. 582, *Economic Growth Center, Yale University*, New Haven, CT, 1989.
- [3] N. G. Shita and L. B. Zeleke, "Predictors of divorce and duration of marriage among first marriage women in Dejene administrative town," *Scientific Reports*, vol. 14, art. 8728, 2024.
- [4] Frank, "Predicting marital stability: An approach for more characteristics," 2024 (framework adopted for the six compatibility dimensions and 0–4 risk scoring).
- [5] International Institute for Population Sciences (IIPS) and ICF, *National Family Health Survey (NFHS), India — Demographic and Health Survey Couples Recode (IACR7EFL)*. (Primary training dataset, ≈46,000–48,000 couples.)
- [6] Love-Marriage / Arranged-Marriage structured dataset (project dataset used to train and serialise the deployable XGBoost and LightGBM compatibility models).
- [7] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [9] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [10] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [11] G. Lemaître, F. Nogueira, and C. K. Aridas, "Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning," *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017. (Provides SMOTETomek and random over-sampling.)
- [12] W. McKinney, "Data structures for statistical computing in Python," in *Proc. 9th Python in Science Conf.*, 2010, pp. 56–61. (pandas.)
- [13] C. R. Harris et al., "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.
- [14] P. Virtanen et al., "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020. (Used for t-test and chi-square validation.)